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PALEOMAGNETISM IN GREECE: INDICATIONS FOR RELATIVE BLOCK MOVEMENT

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ABSTRACT

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There is a difference of 120° between the strike of the Pindos mountain chain and that of the Argolis peninsula. Both consist of rocks of the same age (Triassic Jurassic).

Samples were collected to see if paleomagnetic data also exhibited this difference in angle. 23 samples from two sites and four lava strata of the Pindos resulted in normal and reversed directions with a mean direction $D = 334°$, $I = 22°$ with $\alpha_{95} = 9°$, and 24 samples from four sites of the Argolis peninsula in a mean direction of $D = 82°$, $I = 19°$ with $\alpha_{95} = 17°$. This is a declination difference of $D = 108°$. Therefore, a relative rotational block movement with an angle of about 110° could be assumed. The result depends to a great extent on the dip correction of the lava flows.

INTRODUCTION

In the past few years many hypotheses dealing with the circum-Mediterranean mountain systems in terms of plate tectonics have been proposed (Ryan et al., 1970; Mc Kenzie, 1970). Paleogeographic and geophysical observations form the basis of new interpretations of geodynamic mechanisms and orogenetic phases. In the case of Greece, the Argolis peninsula is located on the southern extension of the Pindos mountain-chain structural line and moreover, the same tectonic and paleogeographic structures occur at the Argolis but with a shift through an angle of 100—120° (Bannert and Bender, 1968). It was felt desirable to test whether this angle could be obtained using an independent method.

GEOLOGICAL SITUATION

The geologic map of Greece shows that the Pindos mountain chain strikes approximately NNW—SSE. Paleogeographic studies by Brunn (1956) and Auboin (1959) demonstrated orogenetic movements and flysch transgression from east to west.

In the Argolis peninsula the strike of the tectonic units is E—W. Paleogeo-

graphic studies showed that the flysch transgression moved from the south to the north (Bannert and Bender, 1968). Consequently it is exceptionally difficult to reconstruct the paleogeography of Greece as a whole, especially considering the migration of the flysch troughs. Other E—W striking tectonic structures are known in Boeotia and Attica. However, the Argolis peninsula is bounded by the Aegaean Sea with various volcanic activities in the past to the south and east, by the tectonically active Corinthian graben with subvolcanic activity to the north, and by a region of Tertiary and Quaternary sediments to the west, deposited after a morphologic depression. East of this depression — the zone of Argos—Corinth — E—W striking tectonic units occur. To the west of it the Peloponnesos tectonic unit strikes north-northeast.

If one rotates the Argolis peninsula 100° anticlockwise, then the tectonic and paleogeographic structures would fit with the major units of Greece.

NOTES ON THE SAMPLE COLLECTING

During the field trip samples were collected in the Pindos mountain chain and on the Argolis peninsula to see whether paleomagnetic measurements supported the hypothesis of an anticlockwise rotation. In the Pindos area the sampling conditions were very bad, partly because of the weathered material of tectonic influence and partly because of bad road conditions. Therefore the authors decided to collect flysch samples as well. But the flysch turned out to be unsuitable for paleomagnetic work because of its sedimentation mechanism. The best paleomagnetic samples were taken from four lava strata of two sites near Mikroleizura, 12 km northeast of Grevena. According to Brunn (1956), this region is built up of Jurassic gabbros which are part of a huge submarine ophiolitic extrusion. Dips of the gabbro flows at the sites are between 215/54 and 232/42 (azimuth/amount). Other samples were taken on the Argolis peninsula at various locations south of Lighourion. They were collected from diabases of Jurassic age belonging to the same volcanic cycle as the gabbros near Grevena (see Fig. 1).

LABORATORY MEASUREMENTS

The measurement of the vector of the natural remanent magnetization and its directional behaviour during increasing a.c.-demagnetization were used to decide if a sample was suitable for more detailed study. If the sample had either of the following two properties in particular, then it was not accepted: (1) a large irregular directional change of the remanence during increasing demagnetization of a sample; (2) the remanence direction of the samples within a hand specimen scattering over a space angle of more than $\pm 50°$, even after demagnetization.

In this way, the flysch and serpentinite specimens were shown to be unsuitable for paleomagnetic statements. It also turned out that most of the collected pillow diabase samples had been reoriented during initial cooling.

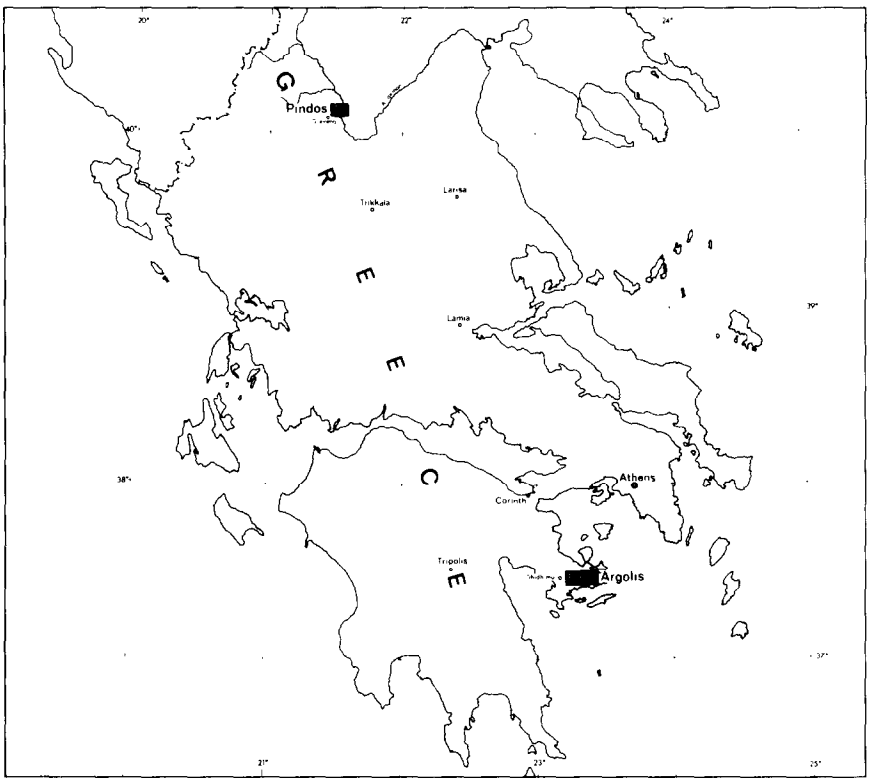


Fig. 1. Map of Greece. Both sampling areas are indicated by cross-hatching and named "Pindos" and "Argolis".

Therefore the authors decided to reject all pillow samples in favor of the compact diabase layers. Using the two criteria mentioned above, about 35% of the sample material from Argolis had to be rejected, by reason of metamorphism, reorientation after partial cooling, fast sedimentation and weathering.

According to the concept of the extrusion of the lavas being more or less horizontal lava flows, the "cleaned" remanence direction was recalculated relative to the strike and dip of the lavas, as a tectonic correction. Fig. 2 shows its effect on the diabase samples from the Argolis peninsula.

Only after dip correction does it become obvious that there are samples of opposite remanence direction. Since it is not known whether the lava layers were originally exactly horizontal or if the axis of the dip correction was always that of the three-dimensional axis of the tectonic unit, the authors estimate a possible error of up to 10° in the remanence direction in either declination and/or inclination.

The fact that both sampling areas include samples of normal and reversed direction (see Fig. 3) is only of marginal interest here. Since at the moment

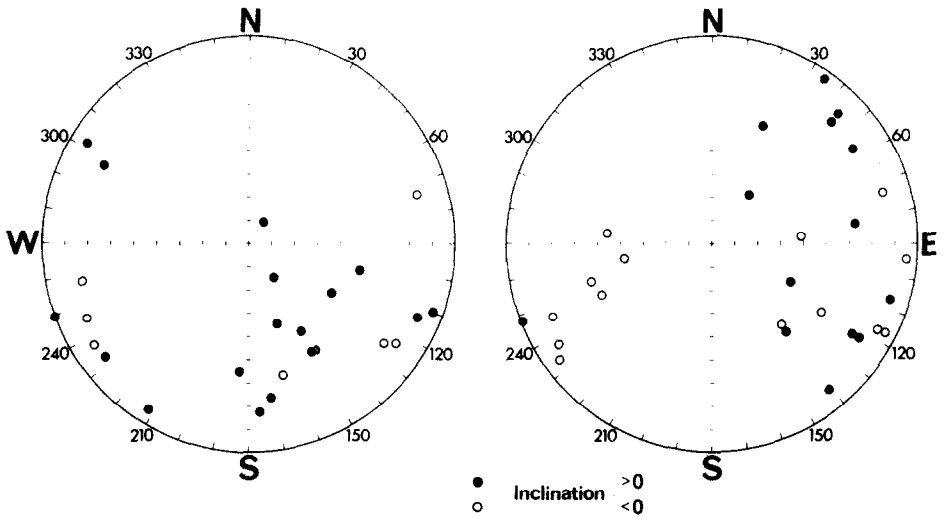


Fig. 2. Equal-area projection of the directions of remanent magnetization of the samples from the Argolis peninsula after a.c.-demagnetization with 500 Oe peak value. Left: without dip correction; right: with dip correction.

only the direction of the remanence axis is of importance, the direction of the "reversed" remanence has been changed into "normal". Then the samples of the Pindos area resulted in a mean direction of the remanence with $D = 334°, I = 22°$ with $\alpha_{95°} = 9°$, and that of the Argolis peninsula in $D = 82°, I = 19°$ with $\alpha_{95°} = 17°$ (see also Table I and Fig. 4). The declination difference of both mean directions appears to be 100—110°.

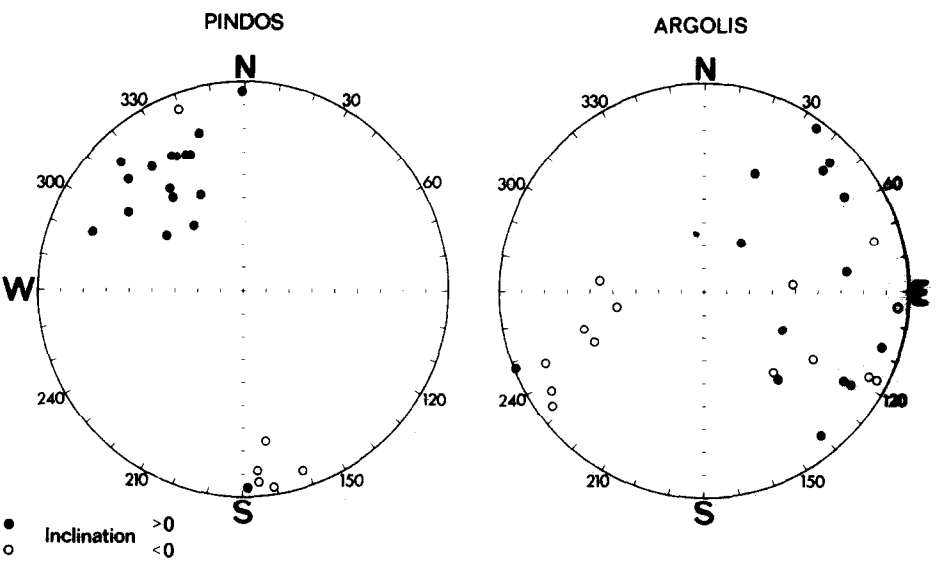


Fig. 3. Equal-area projection of the direction of remanent magnetization of both sampling areas after a.c.-demagnetization (500 Oe) and dip correction.

TABLE I

Data of the magnetic samples

	Pindos area	Argolis peninsula
Type of rock	gabbro	diabase
Sampling sites	2	4
Lava strata	4	not specified
Specimen	7	9
Samples	23	24
Remanence direction	+ and −	+ and −
Natural remanent magnetization (NRM)	122 γ	222 γ
Susceptibility	$1.2 \cdot 10^{-4}$	$11.9 \cdot 10^{-4}$
Remanent magnetization $H_{\sim} = 400$ Oe	21.1γ	18.2γ
Mean-site, dipcorrected		
Declination D	334°	82°
Inclination I	22°	19°
Fisher's statistics α_{95}°	9°	17°
Virtual pole position	112°W 53°N	110°W 12°N

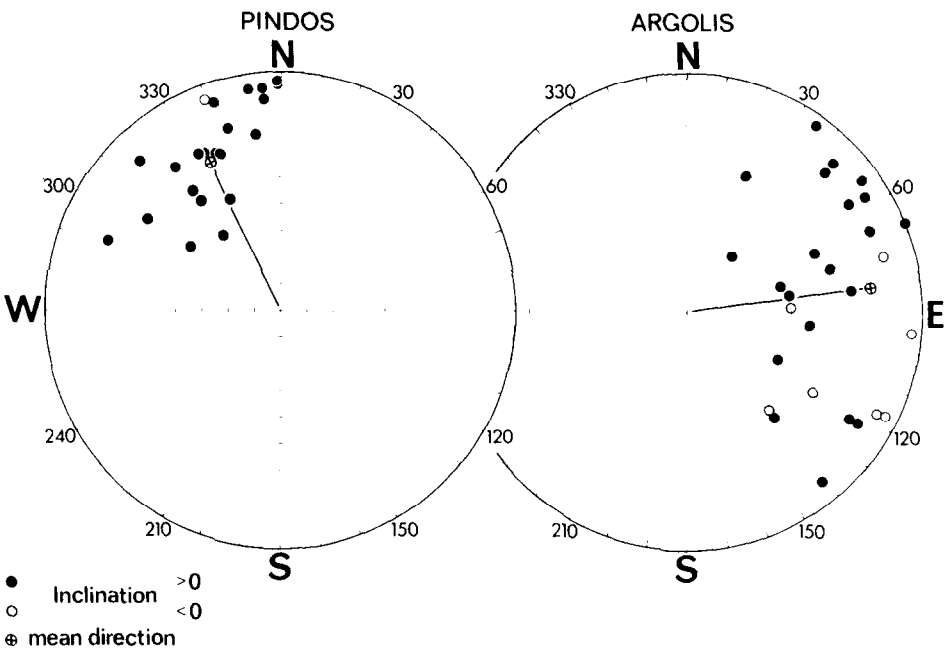


Fig. 4. Projection of reversed magnetized samples of Fig. 3 to "normal" directions. The declination difference of both mean directions is $\Delta D = 108°$.

In order to check the possibility that the remarkable declination difference in both sampling areas was caused by an anisotropic effect of the magnetic material within the rock, measurements of the saturation remanence for pilot samples were made. For comparison the measurements of the three orthogonal components of each sample were normalized to 100%. The bases of comparison were: (1) the scatter of repeated experiments for the same component of one sample; and (2) the deviation of the components of one sample within a measurement. Two samples of different basaltic outcrops were included in this comparison.

(1) From Table II it can be seen that the scatter of repeated experiments is up to 5.4%. At this point there is no remarkable difference between the submarine lava flows and the Tertiary basaltic outcrops.

(2) The saturation remanence of different components of one sample differs between 4.3% and 10.5% and is slightly higher for the samples of the submarine extrusions than for those of the Tertiary basalts.

TABLE II

Normalized saturation remanence of pilot samples for the X, Y and Z components

	Measurements (%)		Maximum difference (%)	
	1	2	between components within one measurement of one sample	of one component between two measurements
Pindos				
Pilot samples 1	100.0	100.0	8.0	5.4
	92.0	92.8		
	98.8	93.4		
2	100.0	100.0	10.5	3.2
	89.5	92.7		
	92.0	93.3		
Argolis				
Pilot samples 1	96.8	93.8	6.0	3.8
	97.8	94.8		
	100.0	100.0		
2	95.7	95.7	4.3	0.0
	100.0	100.0		
	98.6	98.6		
Basalt				
Lamsberg	98.7	95.4	5.7	3.4
	100.0	100.0		
	94.3	96.3		
Basalt				
Blaue Kuppe	100.0	100.0	4.5	2.5
	98.5	96.0		
	95.5	97.6		

These higher differences are due to the special cooling conditions of the submarine lava strata. But the maximum difference of 7—10% in our measurements would have only a small directional influence on the original thermal remanent magnetization. In this respect the declination difference of 108° of the Argolis and Pindos area could not have been caused by an anisotropy of the magnetic material. Neither can it have had its origin in local tectonic causes, since the originally horizontal lava flows have now a dip of about 40—50°.

DISCUSSION OF THE RESULTS

The magnetic data were insufficient to say that the Argolis peninsula definitely rotated relative to the northern larger part of Greece. First the number of samples is statistically too small, secondly the scatter of the Argolis samples is considerable and lastly the distance between the compared sampling areas is large (Fig. 1). The difficulty during the field trip was to find sites with suitable sample material in areas already geologically investigated.

The smaller the value of a parameter (angle difference, intensity, etc.) is, the larger the number of samples or data points must be. In our case the angle in question (100—120°), should be, and is, easy enough to recognise without the help of statistics. Fig. 4 shows a clear difference in declination. The fact that the data include both normal and reversed remanence directions, increases the reliability of the results since it means that the remanence direction was fairly constant for long enough to include at least one reversal and so it excludes even local tectonic movements during the deposition of the lava strata with opposite remanence direction.

Regarding the large distance between the sampling areas, one can say that despite the occurrence of local flysch transgression varying in direction, there is a general directional tendency. The dip and strike at the sampling sites in the Pindos area are comparable to those measured throughout the area; on the Argolis peninsula the data are from samples from four different sites several kilometers apart.

Therefore, the authors assume that the results obtained might be of more than just local importance, and they take the data to be an indication of clockwise rotation of about 100° of the Argolis peninsula relative to the Pindos mountain chain later than Late Jurassic. It is not necessary that the center of an assumed rotation lies outside the peninsula, i.e., a hinge. A center of rotation could be assumed within the peninsula.

From the mean directions of the remanence and their circle of 95% probability of 9° and 17°, respectively, the corresponding virtual pole positions were computed using Fisher's α_{95}° ellipse (Fig. 5). In Fig. 5 the Jurassic pole position for Europe (J_E) and for Africa (J_A) (Zijderveld and Van der Voo, 1973) were plotted. It appears that the pole position of the Pindos area agrees fairly well with that of the African plate. This fact is of great interest, since Turkey also shows the same position with respect to Africa for a long time (Van der Voo, 1968).

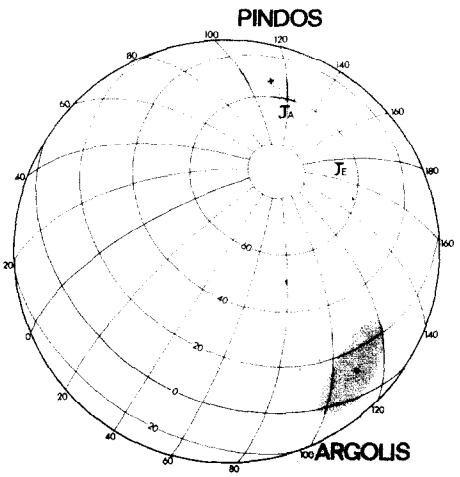


Fig. 5. Virtual pole positions computed from the mean directions of the "Argolis" and "Pindos" samples with Fisher's α_{95}° ellipse (hatched). J_A is the Jurassic pole position for Africa, J_E for Europe.

It should be stressed that in this context paleomagnetic results should not be used on their own, because they depend on the geological age of the samples being correct as well as on the mechanism of the lava deposition here so that the dip reduction is correct. Therefore, this type of problem is at present almost impossible to solve using only one method. With regard to the present study, the authors feel that paleomagnetic studies should be carried out on other E—W striking geological formations, such as those in Boeotia and Attica as well as in the Othrys mountains, in order to investigate this tectonic concept of Greece in more detail.

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